

Day 1 - Data Preparation and Feature Engineering

GSK Medicine Outcome Prediction System

Goal: examine, clean and prepare the clinical dataset prior to machine-learning model development. This report summarises the Day 1 activities and verified outputs.

Stage	Rows	Columns
Raw Data	1,155,000	41
Cleaned Data	1,050,000	31
Engineered Data	1,050,000	38

What We Completed

- 1. Exploratory Data Analysis** — Reviewed structure, missingness, duplicates, invalid clinical values, distributions, correlations and outcome balance.
- 2. Data Cleaning** — Standardised data, removed repeated patient records, corrected inconsistent values, handled missing predictors and capped selected outliers.
- 3. Feature Engineering** — Created seven clinically useful features to enrich model inputs.

Key EDA Findings

- The original dataset contained 1,155,000 rows and 41 columns.
- Several columns represented the same information using inconsistent names.
- Missing values appeared in demographic, clinical and treatment fields.
- 70,017 exact duplicate rows were identified.
- 105,000 additional repeated patient rows were identified.
- 34,983 repeated rows were not completely identical.
- Invalid and extreme clinical values were found and addressed during cleaning.
- Treatment outcomes were imbalanced and some outcomes were missing.

Data Cleaning Performed

- Combined duplicate age, gender and drug-name columns.
- Converted weight from pounds to kilograms where required.
- Extracted systolic and diastolic blood pressure.
- Removed redundant and unusable columns.
- Standardised column names and categorical values.
- Corrected common drug-name and diagnosis variations.
- Parsed dosage, duration and admission dates.
- Converted outcome fields to consistent binary values.
- Replaced impossible clinical values with missing, then imputed predictors.
- Capped selected outliers using IQR.
- Retained one record per unique patient.

Cleaning Result

Final rows: 1,050,000

Final columns: 31

Patient IDs unique: Yes

Feature Engineering

A concise summary of the seven engineered features is provided below.

Feature	Meaning
kidney_stage	Groups patients by kidney function using eGFR
bmi_category	Classifies BMI as underweight, normal, overweight or obese
age_group	Groups patients into clinically meaningful age ranges
liver_risk	Flags elevated ALT or AST liver enzymes
polypharmacy	Flags patients taking five or more concurrent drugs
elderly_high_dose	Flags older patients receiving above-median dosage
de_ritis_ratio	Calculates the AST-to-ALT liver enzyme ratio

Validation Completed

- Input and output row counts were checked.
- Cleaned dataset contains exactly 31 columns.
- Engineered dataset contains exactly 38 columns.
- Patient IDs remain unique after processing.
- All seven engineered features were created successfully.
- The original raw dataset was preserved.

Important Note: 64,124 rows have missing treatment_outcome values. These rows were preserved during preparation but will be excluded before supervised model training.

Next Steps

- Prepare modelling data
- Split into training, validation and testing sets
- Fit encoding and scaling using training data only
- Train and evaluate classification models

Prepared by: Data Engineering Team